

# BAAZ & UDAY JKBOSE

2026 CLASS 11<sup>TH</sup>

## CELL, The Unit of Life

LECTURE-5

ZOOLOGY

DAVOOD SIR



# Topics To Be Covered

- 1 Plastid & its Types
- 2 Ribosome and its types
- 3 Previous Year NEET Questions

# Eukaryotic Cell



# Plastid

↓  
Discovery → Ernst Haeckel.

Named by → " "

Presence → plant cell, Protists like Euglena

# Plastid

Chloroplast (Green coloured).

Green pigment like  
 { Chlorophyll }  
 { Carotenoid }

Responsible for  
 Absorption of  
 Sunlight Energy  
 during photosynthesis.

CHROMOPLAST (Coloured Plastid)

Coloured plastid due to

{ Carotenes }  
 { Xanthophylls }

these pigments gives  
 the part of the plant  
 a yellow, orange & Red  
 colour.

Leucoplast (Colourless plastid).

colourless plastid

Amyloplast

Stores carbohydrate  
 e.g potato

Aleuroplast

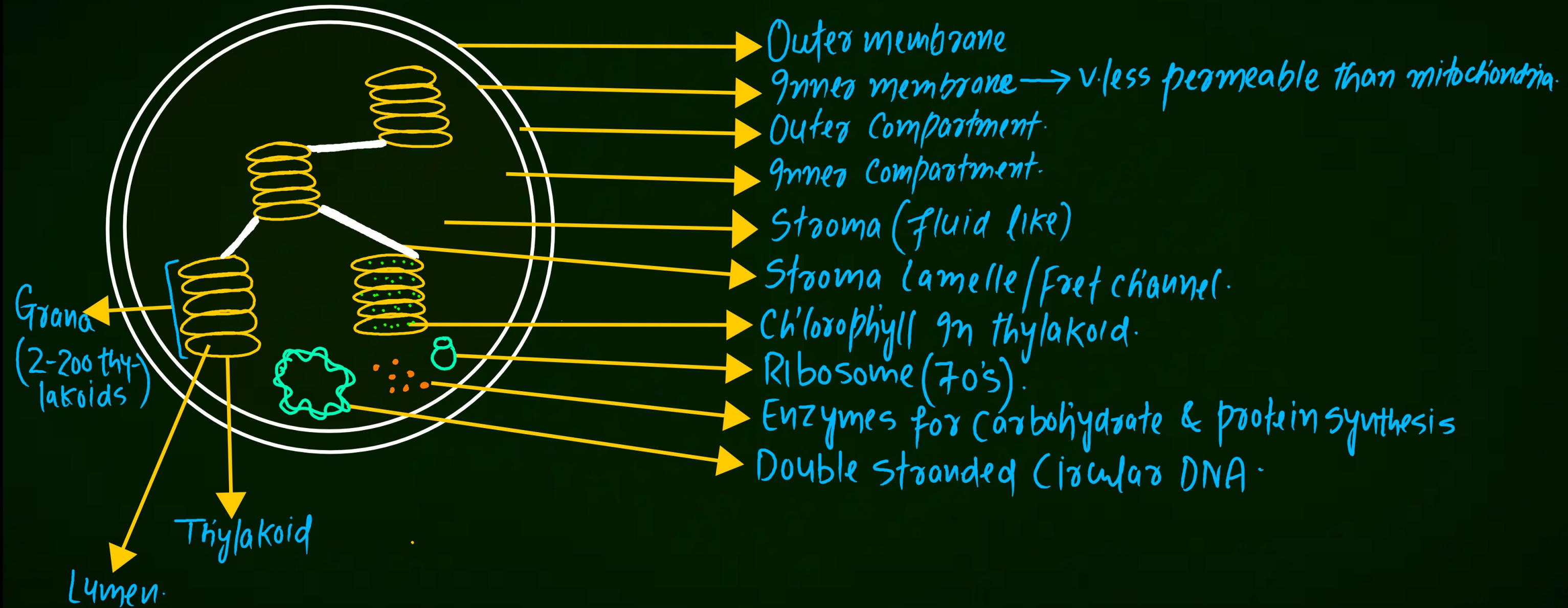
Stores protein  
 e.g ?

Elaioplast

Stores fats

# Structure of a Chloroplast

Chloroplast + Mitochondria = Semi-autonomous cell organelles.



# Plastid

Plastid was discovered by **Earnst Haeckel** in 1866 term Plastid was coined by Earnst Haeckel also. Plastid is a cell organelle found **only in plant cells**, Some **protists like Euglena** and are absent in animal cell.

Based on the type of pigments plastids can be classified into **chloroplasts**, **chromoplasts** and **leucoplasts**.

- The **chloroplasts** contain **chlorophyll and carotenoid pigments** which are responsible for trapping light energy essential for photosynthesis.
- In the chromoplasts **fat soluble carotenoid pigments like carotene, xanthophylls** and others are present. This gives the part of the plant a **yellow, orange** or **red** colour.

# Plastid

The leucoplasts are the colourless plastids of varied shapes and sizes with stored nutrients and are of three types:

- Amyloplasts: Amyloplasts store carbohydrates (starch), e.g, potato.
- Elaioplasts: Elaioplasts store oils and fats.
- Aleuroplasts: Aleuroplasts store proteins.

## Structure of a Chloroplast

**Shape:-** Lens shaped, Oval, spherical discoid or even ribbon like.

**Size:-** 5 to 10micrometer length and 2 to 4 micrometer width.

**Number:-** 1/cell (Chlamydomonas and green algae), 20-40/cell in mesophyll of leaves.

Like mitochondria, the chloroplasts are also double membrane bound. The inner chloroplast membrane is **less permeable**. The space limited by the inner membrane of the chloroplast is filled with the **stroma**. A number of organised flattened membranous sacs called the thylakoids, are present in the stroma. Thylakoids are arranged in stacks like the piles of coins called **grana**. There are also flat membranous tubules called the **stroma lamellae** connecting the thylakoids of the different grana. The membrane of the thylakoids enclose a space called a **lumen** in which chlorophyll pigments are present. The stroma of the chloroplast contains **enzymes required for the synthesis of carbohydrates and proteins**. It also contains small, **double stranded circular DNA molecules and ribosomes (70S type)**.

**Question: NEET-2024**

4 Marks.



**The DNA present in chloroplast is**

**1** linear, double stranded ☒

**2** circular, double stranded ☒

**3** linear, single stranded ☒

**4** circular, single stranded. ☒



**Question: NEET-2024**

4 marks

Given below are two statements:

**Statements I:** Mitochondria and chloroplast both double membrane bound organelles. T.

**Statements II:** Inner membrane of mitochondria is relative less permeable, as compared chloroplast. F.

In the light of the above statements, choose the appropriate answer from the options given below.

- 1 Both Statement I and Statement II are correct.
- 3 Both Statement I and Statement II incorrect.
- 2 Statement I is correct but Statement II incorrect. ✓
- 4 Statement I is incorrect but Statement II incorrect.

# Question: NEET-2021

Match List – I with List – II.

**List – I**

- (p) Cristae → (iii)
- (q) Thylakoids (iv)
- (r) Centromere (i)
- (s) Cisternae (ii)

**List – II**

- (i) Primary constriction in chromosome
- (ii) Disc-shaped sacs in Golgi apparatus
- (iii) Infoldings in mitochondria
- (iv) Flattened membranous sacs in stroma of plastids

Choose the correct answer from the options given below

- ~~1~~ (p)-(ii), (q)-(iii), (r)-(iv), (s)-(i)
- ~~2~~ (p)-(iv), (q)-(iii), (r)-(ii), (s)-(i)
- ~~3~~ (p)-(i), (q)-(iv), (r)-(iii), (s)-(ii)
- 4 (p)-(iii), (q)-(iv), (r)-(i), (s)-(ii)

**Thank**  
*me*